FOR A CATEGORY WITH CHAOTIC TOPOLOGY AND FLAT RESTRICTION MAPS, QUASI-COHERENT OBJECTS OF THE DERIVED CATEGORY ARE EXACTLY THE COMPLEXES WITH QUASI-COHERENT COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the definition of D_QCoh(O) under the assumed flatness, the four-term computation of H^i(M)(U) for M quasi-coherent, and the converse obtained from the module-level criterion for quasi-coherent modules on a chaotic site.

Editorial note. The label lemma-cartesian-flat-quasi-coherent names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Chaotic topology and the identification of sections with values on objects. Same chapter, the chaotic topology on a category C and the exactness of taking sections over an object of C. Supplies the exactness of M |-> M(U) which yields the first and last equalities in the computation of H^i(M)(U).
- Weak Serre property of the quasi-coherent modules and the module-level description. Modules on Sites, the lemmas stating that QCoh(O) is a weak Serre subcategory under the flatness assumption and characterising quasi-coherent modules on a chaotic site. Supplies the characterisation of quasi-coherence used in both directions of the proof.
- Derived category of a weak Serre subcategory. Derived Categories, the section on triangulated subcategories, used to give meaning to D_QCoh(O). Supplies the definition of the full subcategory D_QCoh(O) appearing in the statement.

Source reference. Original section 06UP: sources/stacks-chow/tags.txt, source lines 8753–8753. Prerequisite pointer invoked at proof line(s) 11572; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GZP: sources/stacks-chow/tags.txt, source lines 21184–21184. Prerequisite pointer invoked at proof line(s) 11566; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0GZN: sources/stacks-chow/tags.txt, source lines 21183–21183. Prerequisite pointer invoked at proof line(s) 11591; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let $\mathcal{C}$ be a category viewed as a site with the chaotic topology. Let $\mathcal{O}$ be a sheaf of rings on $\mathcal{C}$. Assume for all $U \to V$ in $\mathcal{C}$ the restriction map $\mathcal{O}(V) \to \mathcal{O}(U)$ is a flat ring map. Then $QC(\mathcal{O})$ agrees with the subcategory $D_{QCoh}(\mathcal{O}) \subset D(\mathcal{O})$ of complexes whose cohomology sheaves are quasi-coherent.*

Proof. Recall that $QCoh(\mathcal{O}) \subset Mod(\mathcal{O})$ is a weak Serre subcategory under our assumptions, see Modules on Sites, Lemma 0GZP. Thus taking the full subcategory

$$D_{QCoh}(\mathcal{O}) = D_{QCoh(\mathcal{O})}(Mod(\mathcal{O}))$$

of $D(\mathcal{O})$ makes sense, see Derived Categories, Section 06UP. (Strictly speaking we don't need this in the proof of the lemma.)

Let M be an object of $QC(\mathcal{O})$. Since for every morphism $U \to V$ in $\mathcal{C}$ the restriction map $\mathcal{O}(V) \to \mathcal{O}(U)$ is flat, we see that

$$\begin{aligned} H^i(M)(U) &= H^i(R\Gamma(U, M)) \\ &= H^i(R\Gamma(V, M) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U)) \\ &= H^i(R\Gamma(V, M)) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) \\ &= H^i(M)(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U) \end{aligned}$$

and hence $H^i(M)$ is quasi-coherent by Modules on Sites, Lemma 0GZN. The first and last equality above follow from the fact that taking sections over an object of $\mathcal{C}$ is an exact functor due to the fact that the topology on $\mathcal{C}$ is chaotic.

Conversely, if M is an object of $D_{QCoh}(\mathcal{O})$, then due to Modules on Sites, Lemma 0GZN we see that the map $R\Gamma(V, M) \to R\Gamma(U, M)$ induces isomorphisms $H^i(M)(U) \to H^i(M)(V) \otimes_{\mathcal{O}(V)} \mathcal{O}(U)$. Whence $R\Gamma(V, K) \otimes^{\mathbf{L}}_{\mathcal{O}(V)} \mathcal{O}(U) \to R\Gamma(U, K)$ is an isomorphism in $D(\mathcal{O}(U))$ by the flatness of $\mathcal{O}(V) \to \mathcal{O}(U)$ and we conclude that M is in $QC(\mathcal{O})$. □